Solutions to Chapter 3 Exercises (Sutton & Barto)

Aniket Mandal

Exercise 3.1: Three Example Tasks in the MDP Framework

Example 1: Autonomous Delivery Drone

- **States:** Drone's location, battery level, weather conditions.
- **Actions:** Move North, South, East, West; Deliver Package; Recharge.
- **Rewards:** +10 for delivery, -1 for movement, -20 for crash.

Example 2: Personalized Learning Assistant

- **States:** Student's skill level, topic understanding.
- **Actions:** Recommend video, quiz, explanation, or break.
- **Rewards:** +5 for improvement, -1 for disengagement, +1 for engagement.

Example 3: Urban Traffic Light Control

- **States:** Vehicle counts, time of day.
- **Actions:** Change phase, extend green, switch red.
- **Rewards:** -1 for congestion, +5 for flow, -10 for accidents.

Exercise 3.2: Limits of the MDP Framework

The MDP framework assumes full observability and stationary environments. It struggles with:

- **Partially Observable Tasks:** e.g., Poker.
- **Non-stationary Goals:** e.g., reacting to rare stock events.

Exercise 3.3: Agent-Environment Boundary in Driving

There's no fixed rule. The correct boundary depends on:

- **Abstraction Level:** High-level decisions (like "drive to office") may simplify learning.
- **Observability:** Use where feedback and control are reliable (e.g., steering, brake).

Exercise 3.4: Table for $p(s', r \mid s, a)$

s	a	s'	r	$p(s', r \mid s, a)$
High	Search	High	1	0.9
High	Search	Low	1	0.1
High	Wait	High	0	1.0
Low	Search	Low	1	0.2
Low	Search	Low	0	0.8
Low	Wait	Low	0	1.0
Low	Recharge	High	0	1.0

Exercise 3.7: Maze Robot and Sparse Reward

If the robot gets +1 for escaping and 0 otherwise, it receives a total reward of 1 no matter how long it takes. There's no feedback to encourage faster escapes.

Fix:

- Penalize each step: e.g., reward = -0.01 per step.
- Use discounting to prefer quicker rewards.

Exercise 3.16: Constant Addition in Episodic Tasks

Adding a constant c to all rewards changes the return in episodic tasks:

$$G'_t = G_t + c(T - t)$$

This biases the agent to prefer longer episodes. **Example:** If agent delays reaching the goal to collect more +0.1 per step, it no longer learns to escape quickly.

Exercise 3.20: Value Sensitivity to Policy

Changing the policy π affects action probabilities $\pi(a|s)$, altering expected returns. Even small changes in π can significantly change $v_\pi(s)$, especially in long-term or high-impact decision tasks.

Exercise 3.21: Same v_π, Different q_π

Two policies can yield the same state-value function:

$$v_\pi(s) = \sum_a \pi(a|s) q_\pi(s, a)$$

but differ in how they distribute probabilities over actions. Hence, their $q_\pi(s, a)$ can be different.

Exercise 3.22: Greedy Policy Not Always Optimal

Even if:

$$\pi(s) = \arg\max_a q_\pi(s, a)$$

that doesn't guarantee $\pi = \pi^*$, since q_π is based on π's behavior, not the optimal return.

Exercise 3.23: Greedy Policies and Equal Value

If $\pi_1(s) = \pi_2(s) = \arg\max_a q(s, a)$, both policies act the same and therefore:

$$v_{\pi_1}(s) = v_{\pi_2}(s) \quad \forall s$$